

4. (a) Limestone quarrying is an important business in the UK. Quarrying creates jobs in areas where there are often limited opportunities. There is a huge demand for the products of quarrying, such as building stone and cement. Good roads and rail links are needed for transporting the products of quarrying. Thousands of people are employed in quarrying and related industries.

Many people argue against the opening of a quarry in their area despite all the benefits of limestone quarrying. Explain **two** reasons local people might use in an attempt to stop the opening of a new quarry in their area. [2]

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- (b) When limestone is heated in a lime kiln thermal decomposition takes place.

Write a balanced **symbol** equation for this reaction. [1]

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- (c) Calcium silicide is a compound containing calcium and silicon only. A sample of calcium silicide was found to contain 2.0 g of calcium and 2.8 g of silicon.

Calculate the simplest formula for this compound. You **must** show your working. [3]

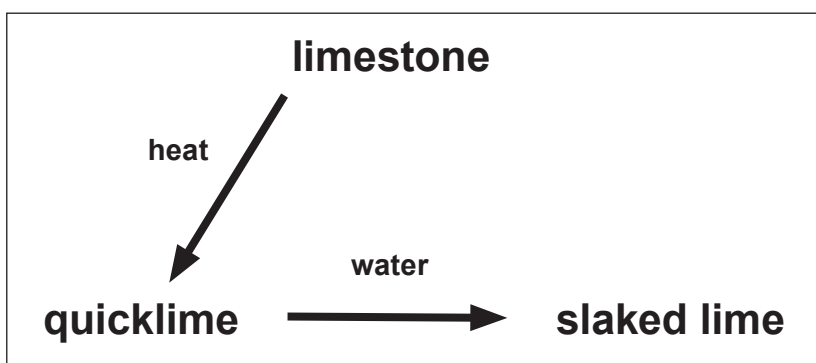
Simplest formula

- Describe and explain the sequence of reactions carried out in the laboratory to convert limestone into slaked lime. [6 QER]

6



2. The following diagram shows how limestone, CaCO_3 , can be converted into useful products.



- (a) When a piece of limestone is heated strongly its mass decreases.

State the type of reaction taking place.

[1]

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- (b) (i) Describe what is **seen** when limestone is heated and converted into quicklime. [1]

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- (ii) Write a balanced symbol equation for the reaction taking place.

[2]

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- (c) (i) Describe what is observed when quicklime is converted into slaked lime. [1]

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- (ii) Write a balanced symbol equation for the reaction taking place.

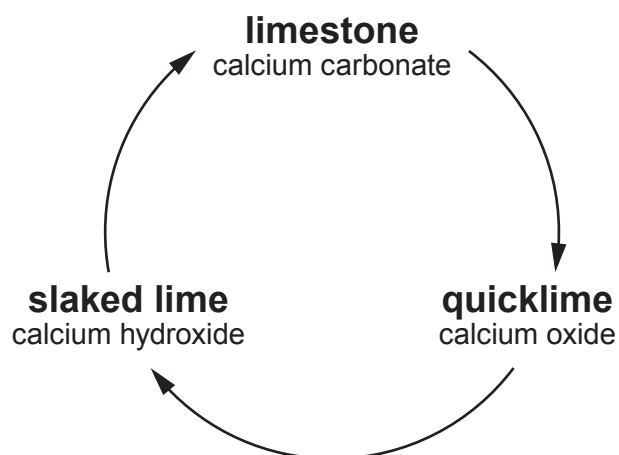
[2]

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Answer **all** questions.

1. Limestone is a rock which consists mostly of calcium carbonate. The diagram shows a cycle of reactions involving limestone.



- (a) (i) When limestone is heated, calcium carbonate is converted to calcium oxide and carbon dioxide.

I. State the name for this type of reaction. [1]

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II. Write a balanced symbol equation for the reaction. [2]

..... → +

(ii) State what must be added to calcium oxide to form calcium hydroxide. [1]

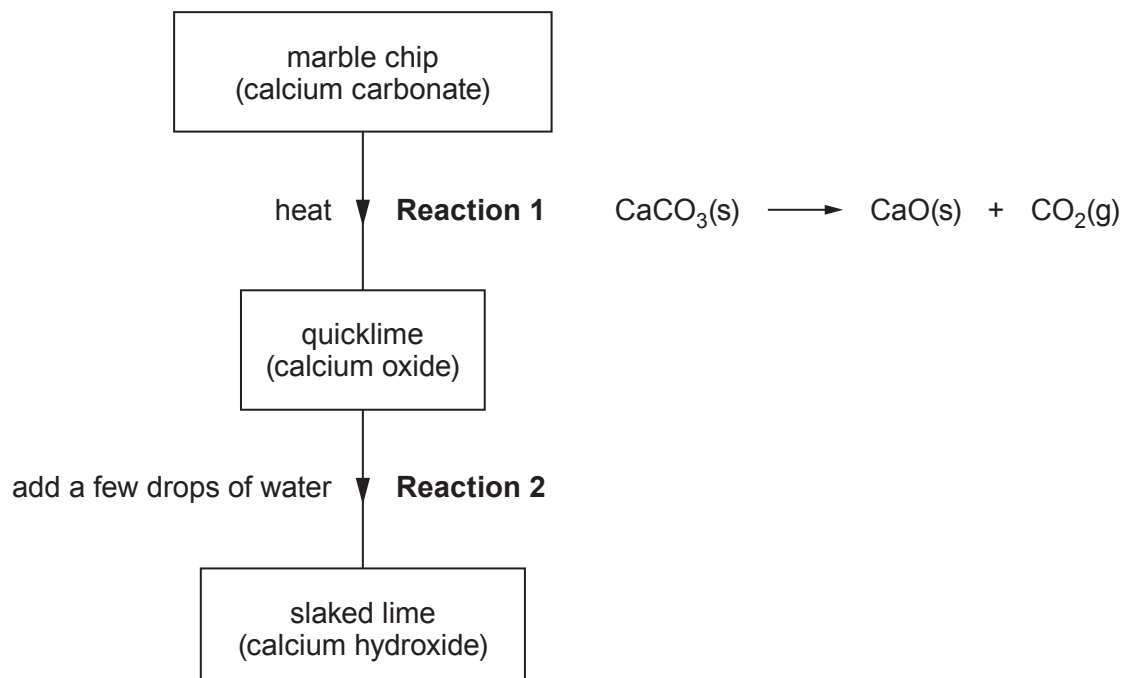
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(b) Give **one** use of limestone in the construction industry. [1]

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4. (a) The flow diagram shows the reactions necessary to make slaked lime (calcium hydroxide) from a marble chip (calcium carbonate).



- (i) In **Reaction 1**, one compound breaks down when heated to form two products.

Complete the name of this type of reaction.

[1]

thermal

- (ii) Give **two** observations you would expect to make as **Reaction 2** happens.

[2]

1

2

- (iii) Complete the symbol equation for the reaction between calcium oxide and water in **Reaction 2**.

[1]

..... + H_2O $\longrightarrow$



- (b) (i) The relative formula mass of calcium hydroxide is 74.

Calculate the number of moles of calcium hydroxide in 2.96 g.

[2]

Number of moles = mol

- (ii) Explain why calcium hydroxide is used to treat some soils.

[2]

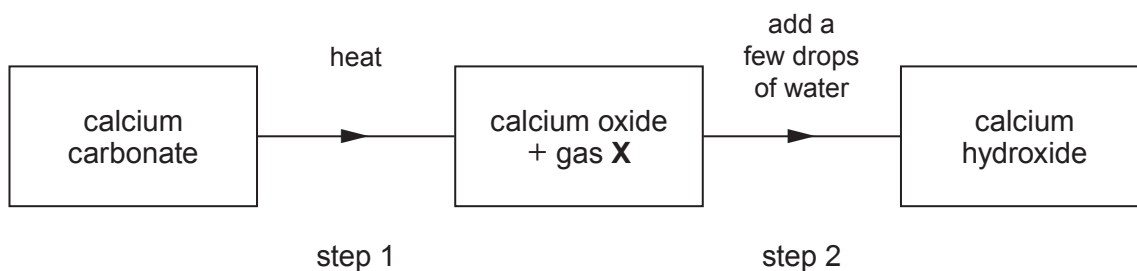
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9. (a) The flow diagram shows the reactions carried out to prepare calcium hydroxide from calcium carbonate, CaCO_3 .



- (i) Name the type of reaction taking place in step 1. [1]

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- (ii) Give the name of gas X. [1]

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- (iii) Give **one** observation that shows that step 2 is exothermic. [1]

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- (iv) Write a balanced equation for the reaction of calcium oxide and water to form calcium hydroxide. [2]

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- (b) Give a reason why no reaction occurs when sodium carbonate is heated with a Bunsen burner. [1]

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- (c) Barium hydroxide is another Group 2 hydroxide.

Solutions of calcium hydroxide and barium hydroxide are both alkalis and turn pH paper purple.

State how you could distinguish between calcium hydroxide and barium hydroxide using a flame test. [1]

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